

Prescribed fire in longleaf pine forests

Longleaf pine forests and plant community

The longleaf pine forest is an ecosystem found in the southeastern United States, and has varied habitat types depending on soils and topography. All longleaf pine habitats are characterized by a widely spaced overstory dominated by longleaf pine, a lack of midstory trees, and an open, prairie-like understory with vigorous ground cover (Oswalt et al. 2012). The flatwood habitat type of the longleaf pine forest has level topography with poorly drained soils and patches of swamp and wet prairie (Habitats). Flatwoods have very high plant diversity in their groundcover plants, and species include orchids and carnivorous plants because of the low nutrient levels (Habitats). The overstory sometimes includes slash pine in addition to longleaf pine (Oswalt et al. 2012). Flatwood longleaf pine forests are generally high productivity and have a tall overstory (Habitats).

The characteristics of longleaf pine forests are maintained by frequent, low severity fires (Oswalt et al. 2012). Fire in this ecosystem creates seedbed conditions needed for the regeneration of longleaf pine seedlings, reduces the density of understory vegetation and creates new areas for herbaceous plants to grow, releases nutrients from dead organic matter into the soil for uptake by plants, improves forage for grazing wildlife, enhances wildlife habitat, excludes invasive plants that are not fire adapted, controls harmful insects and pathogens, and reduces fuel levels and wildfire hazard (Brockway and Lewis 1997).

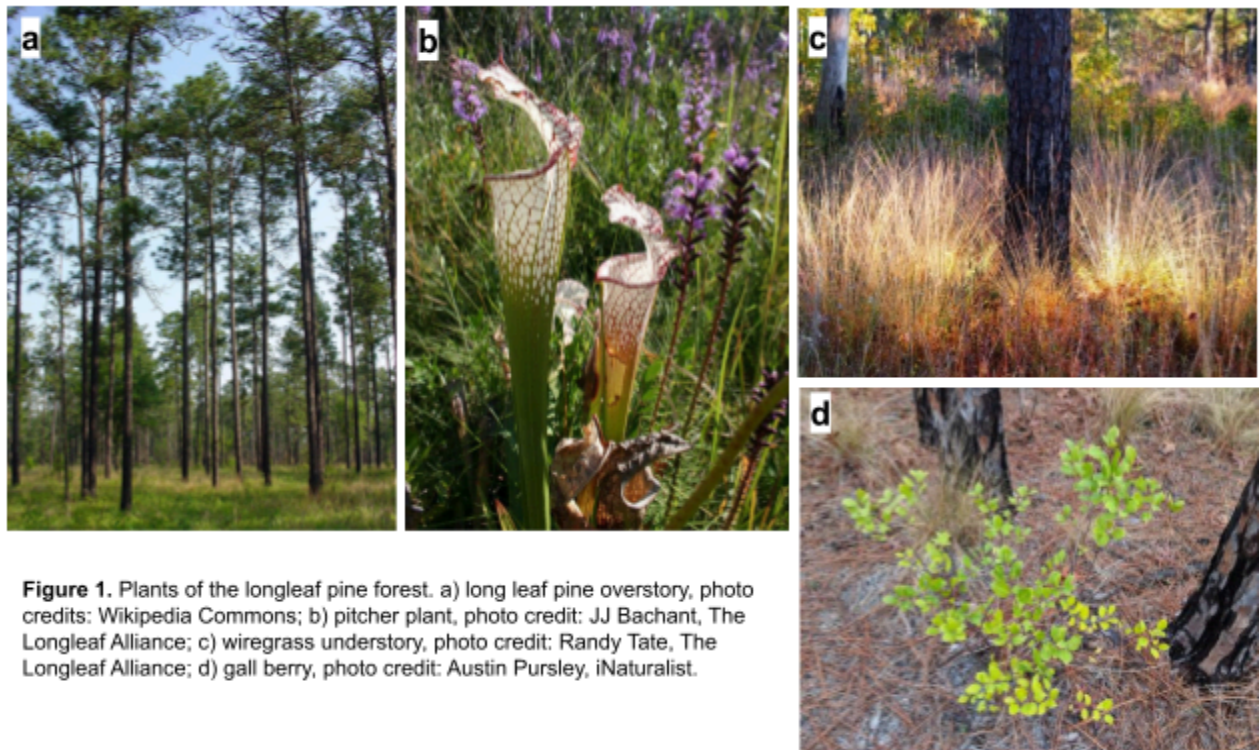


Figure 1. Plants of the longleaf pine forest. a) long leaf pine overstory, photo credits: Wikipedia Commons; b) pitcher plant, photo credit: JJ Bachant, The Longleaf Alliance; c) wiregrass understory, photo credit: Randy Tate, The Longleaf Alliance; d) gall berry, photo credit: Austin Pursley, iNaturalist.

In a hypothetical burn unit in a flatwood longleaf pine forest, the overstory has longleaf pine (*Pinus palustris*) and slash pine (*Pinus elliottii*). There is no midstory layer, but there are shrubs present: gallberry (*Ilex glabra*) is the most common, followed by saw palmetto (*Serenoa repens*), and runner oaks (*Quercus pumila*, *Q. minima*) and blueberries (*Vaccinium myrsinites*, *Gaylussacia dumosa*) appear scattered throughout the stand. The grass portion of the groundcover is mainly wiregrass (*Aristida stricta*) and bluestems (*Andropogon* spp.). Sedges (*Rhynchospora* spp., *Carex* spp.) are also common. Forbs present in the unit are tarflower (*Bejaria racemosa*), hatpins (*Eriocaulon* spp.), pitcher plants (*Sarracenia* spp.), sundews (*Drosera* spp.), and deer tongue (*Carphephorus odoratissimus*). (Glitzenstein et al. 1995; Brockway and Lewis 1997; Habitats)

Table 1. Plant community in example flatwood longleaf pine burn unit

Overstory trees	Shrub layer	Grasses and sedges	Forbs
- Longleaf pine (<i>Pinus palustris</i>) - Slash pine (<i>Pinus elliottii</i>)	- Gallberry (<i>Ilex glabra</i>) - Saw palmetto (<i>Serenoa repens</i>) - Runner oaks (<i>Quercus pumila</i>, <i>Q. minima</i>) - Blueberries (<i>Vaccinium myrsinites</i>, <i>Gaylussacia dumosa</i>)	- Wiregrass (<i>Aristida stricta</i>) - Bluestems (<i>Andropogon</i> spp.) - Sedges (<i>Carex</i> spp.) - Beaksedges (<i>Rhynchospora</i> spp.)	- Tarflower (<i>Bejaria racemosa</i>) - Deer tongue (<i>Carphephorus odoratissimus</i>) - Hatpins (<i>Eriocaulon</i> spp.) - Pitcher plants (<i>Sarracenia</i> spp.) - Sundews (<i>Drosera</i> spp.)

Plant responses to fire

Longleaf pine forests are dominated by fire tolerant plants. Longleaf pine and bunch grasses that make up a majority of the ground cover facilitate fire through their traits, such as dry, fibrous grass blades, and long, thin needles with high amounts of volatile chemicals. These traits allow the forest to undergo frequent low severity fire by creating a continuous fuel bed that quickly grows back post-fire (Brockway and Lewis 1997). Without fire, longleaf pine forests develop a midstory with shrubs and hardwood trees that collect pine needles and create conditions more conducive to severe, stand-replacing fire (Brockway and Lewis 1997).

Overstory

Longleaf pine has distinct life stages that each respond differently to fire. Seeds will germinate on mineral soil exposed by fire. Seedlings that have not reached the grass stage or are early in the grass stage are killed by low severity fire. Once they grow into grass stage seedlings, longleaf pines become very fire resistant because of the dense tuft of needles with high moisture content that protects the bud from high heat. Grass stage seedlings will resprout

from the root collar if killed by fire. Otherwise, fire will stimulate height growth and propel the seedling into the bottlebrush stage. During the bottlebrush stage and early sapling stages, longleaf pines are more susceptible to fire because they have not developed thick bark yet. Fire can cause a reduction in basal area growth or mortality. As saplings grow, they develop thicker bark and are more resistant to fire. These trees may face a reduction in diameter growth if they experience crown scorch. Mature longleaf pines prune their lower branches and have thick bark that flakes off as it burns, reducing heat to the cambium layer. These mature trees are rarely affected by fire except high severity crown fire, which is rare in open longleaf pine stands. (Carey 1992b)

Slash pine is also fire resistant and develops thick, flaky bark and a high, open crown like longleaf pine. Young slash pine develops from a seed to a grass stage seedling and then has rapid height growth. Very young slash pines under two years old are killed by low severity fire, 3-4 year old slash pines are killed by moderate severity fire, and slash pines under 10 years old can survive moderate severity fire. During the grass stage, seedlings killed by fire can resprout from the root collar. After 10-12 years, slash pines are resistant to fires that do not create crown scorch. Fires that cause crown scorch or other damage can reduce growth, but low severity fires with only light crown scorch can enhance slash pine growth. (Carey 1992a)

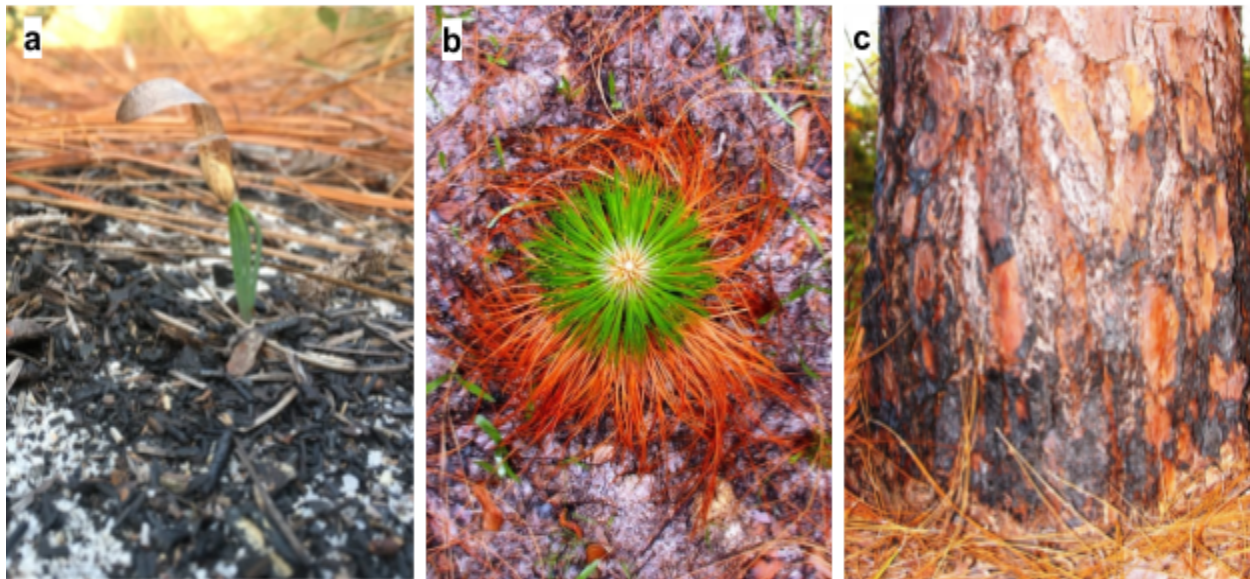


Figure 2. Responses of longleaf pine to fire. a) Longleaf pine seed germinating post-fire, photo credit: Sarah Crate, The Longleaf Alliance; b) New needle growth on grass stage longleaf pine, photo credit: Randy Tate, The Longleaf Alliance; c) Charred bark on a mature longleaf pine, photo credit: Robert Abernethy, The Longleaf Alliance.

Shrubs

Gallberry has very flammable leaves and is usually top-killed by fire. The post-fire regeneration strategy of gallberry is to resprout from rhizomes or seed in from off site by animal dispersers or water. Small gallberries are killed by fire, but large gallberries are able to resprout. (Coladonato 1991)

Saw palmetto is very resilient after fire. Saw palmetto has very flammable foliage and is usually top-killed by fire, but resprouts from root crowns rhizomes except in very high severity fire that kills the rhizomes under the soil organic layer. Saw palmetto resprout soon after fire,

even in dormant season fires. Resprouting after a burn uses carbohydrate reserves, so flowering and fruiting is less common after fire for a year or more. (Van Deelen 1991)

Like the other shrubs in longleaf pine forests, blueberries are rhizomatous. Fire usually top-kills blueberries, but they are able to survive and resprout vigorously from rhizomes. Blueberries may exist as small shrubs with very very low aboveground biomass, and then rapidly grow after a fire. (Tirmenstein 1990)

Groundcover

The most common groundcover species in longleaf pine forests are sun loving perennials with the ability to resprout after fire (Oswalt et al. 2012). Wiregrass is highly flammable because of its thin, fibrous leaves and high percentage of dead leaves. Fire usually consumes all above ground biomass, but the meristems of the plant exist 1-1.5 inches below the surface of the soil, so they are not affected by fires that pass quickly. After fire, the meristems create new top-growth that grows rapidly and rebound to pre-fire levels after a month. Dormant season burns generally do not result in flowering, but growing season fires can cause flowering. (Uchytel 1992b)

Similarly to wiregrass, bluestems are top-killed by fire, but regrow from belowground meristems or basal culm buds after fire and recover quickly. (Walsh 1994; Uchytel 1992a).

Pitcher plants are usually top-killed by fire and survive by belowground rhizomes. Moderate severity fire stimulates growth by releasing nutrients into the soil, removing old growth, and reducing competing vegetation. High severity fire can kill pitcher plants. (Walkup 1991 Oct; Walkup 1991 Nov).

Sundews are sensitive to fire, but occur in habitats that are maintained by frequent fire. Sundews are usually killed by even low severity fire, but survive in unburnt patches and recolonize burnt areas by seeding in (Matthews 1994).

Implications of burn timing

Longleaf pine forests evolved with variable frequency of growing-season fires from lightning strikes, but management of longleaf pine forests utilizes prescribed fire during all seasons (Brockway and Lewis 1997). There is not a strong influence of season of burn on overall dynamics of longleaf pine forests, but some effect exists for longleaf pine seedlings and understory vegetation. Longleaf pines past the seedlings stage do not show any clear trends related to burn timing in flatwood forests (Glitzenstein et al. 1995; Haywood 2012; Willis et al. 2021). Mature overstory longleaf pines in a long-term experiment in southern Alabama found that survival was highest after spring burns, but was not significantly different from any other season (Willis et al. 2021). All average survival rates were 79% or higher, which demonstrates that mature longleaf pines are resistant to burns throughout the year. Longleaf pine seedlings, however, are more sensitive to differences in burn season. Seedling survival is lower after high intensity summer burns than after high intensity winter, fall, and spring burns (Brethauer et al. 2021). This is because seedlings have higher starch reserves outside of the growing season, so they can regrow lost foliage and don't have to rely on photosynthesis for an energy source (Brethauer et al. 2021). Seedlings are more likely to start their height growth into the bottle brush stage after late spring or summer fires than winter fires (Carey 1992b).

Burn seasonality does have some influence on the understory of longleaf pine forests. Seed production by wiregrass and other understory species is stimulated by burning during the growing season that more closely resembles the natural disturbance regime (Brockway and Lewis 1997). Dormant season burns may increase the amount of woody understory plants in some areas because they stimulate resprouting, while growing season burns increase the cover of grass and forb species and lead to fewer and smaller understory trees and shrubs (Haywood 2012). This is because growing season burns expose more mineral soil that is necessary for germination for many understory species and longleaf pine, and they are more effective at reducing woody plant cover because hardwoods have fewer root reserves (Carey 1992b).

Table 2. Implications of burn timing by season

Season	Growing season	Dormant season
Implications	- Expose more mineral soil for seed germination - Kill more woody plants, woody plants grow back sparser, smaller - Increase pine seedling mortality in high severity fire - Stimulate height growth in grass stage pines - Cause flowering in some groundcover species - High temperatures, low wind speed, and high mixing height limit burn windows	- Reduce woody species cover, but stimulate resprouting of woody species - Reduce mortality of pine seedlings in high severity fire - Stimulate height growth in grass stage pines - May not burn through wetland areas, which overtime degrades habitat for salamanders - Time nutrient pulses for high forest productivity - Low relative humidity and high wind speed limit burn windows

Implications of burn conditions (fuel moisture, ambient temperature, wind characteristics)

Burn conditions for prescribed fire are important for achieving burn objectives, keeping the fire contained, and safety of crew members. Fires in longleaf pine forests are affected by fuel moisture, temperature, and wind conditions similarly to other forest types. Fuel moisture has a strong influence on fuel consumption in longleaf pine forests. Moist conditions in litter and duff lead to very low levels of consumption of duff and only moderate amounts of litter consumption (Ferguson et al. 2002). Drier conditions lead to high duff and almost complete litter consumption (Ferguson et al. 2002). Additionally, relative humidity has a suppression effect on fire behavior because of its influence of fuel moisture. Burns that start only shortly before afternoon increases in relative humidity experienced less fire behavior than burns that start earlier when relative humidity is lower (Ferguson et al. 2002).

Wind has a large impact on fire rate of spread, flow of heat energy, and residence time. During a burn at low wind speeds, fire rate of spread increases slightly with increasing wind speed, but after wind speed reaches about 10 m/s, the fire rate of spread increases considerably with each unit increase in wind speed (Butler et al. 2019). Furthermore, fire burning during low wind speeds transfers heat mainly through radiative energy transport, and radiative heat transfer increases with increasing wind speed. Convective heat transfer is more

important for energy transport at higher wind speeds, and increases more slowly with increasing wind speed than radiative heat transfer (Butler et al. 2019). Another important fire behavior variable influenced by wind speed is residence time. Longer residence times happen under the lowest wind speed conditions, and short residence times occur when wind speeds are the highest (Butler et al. 2019). Depending on site conditions and the corresponding objectives of the burn, wind speed can be used to increase or decrease residence time, rate of spread, and radiative and convective heating.

The burn conditions that affect if a burn manager will conduct a prescribed fire depend largely on the intended season of the burn. Drought and days since rain are important for burn managers to account for across all seasons, but temperature related variables are more important during growing season burns (Kupfer et al. 2022). High temperatures are a limitation for prescribed fire practitioners because high temperatures can increase fire intensity and flame lengths which make fires harder to control and more likely to cause unwanted tree damage (Kupfer et al. 2022). For similar reasons, low relative humidity and high wind speeds are a large concern for dormant season fires because the southeast experiences relatively dry winters that increase fire behavior (Kupfer et al. 2022). On the other hand, low wind speeds in the growing season are a constraint on burning because they make fire less predictable, especially on the level topography of flatwood longleaf pine forests (Kupfer et al. 2022). Mixing height is also more of a concern during the growing season because the mixing layer is usually higher and can make fire behavior erratic, impacting crew safety and ability to follow burn plans to prescription (Chiodi et al. 2018).



Figure 3. Prescribed burn being conducted by The Nature Conservancy to restore longleaf pine habitat in Louisiana, photo credit: © The Nature Conservancy

Potential interactions among species following the burn.

After prescribed fire, the density of residual mature longleaf pine trees affects regeneration of grass-stage seedlings. Forests with low mature tree density facilitate higher seedling success because longleaf pine is shade intolerant (Glitzenstein et al. 1995; Zampieri and Pau 2022). However, if tree density is very low, or fire creates large gaps, increased sun exposure and moisture stress can cause seedling mortality, and dispersal distance may be too far for successful regeneration (Zampieri and Pau 2022). Seedling survival after fire is also influenced by the presence of oaks. Although oaks can create competition for longleaf pine seedlings, at low covers created by frequent fire, they facilitate longleaf pine seedling establishment by creating suitable microhabitats (Zampieri and Pau 2022). Similarly, groundcover plants are also influenced by competition post-fire. Shrub resprouting created by dormant season burns can decrease cover of grasses because they create shade and can smother lower plants with litter (Brockway and Lewis 1997). Additionally, understory plants compete with pine seedlings because they have the ability to rapidly colonize root gaps in the soil after disturbance (Pecot et al. 2007).

For wildlife, attributes of forest structure created by repeated fire are important for habitat. Moderate fire frequencies of 3-5 years support the greatest diversity of vertebrate wildlife species because they create a wider variety of forest conditions (Darracq et al. 2016). Habitat created by a range of vegetation heights is not as common in high fire frequency forests, and may have a negative effect on wildlife diversity. Multiple bird species found in longleaf pine forests prefer nesting or foraging in the midstory cover that grows in less-frequently burned forests. Mast for food for squirrels, birds, and deer, and a moist litter layer for a variety of snakes are other important environmental components that develop in a moderate fire frequency (Darracq et al. 2016). Despite this, high fire frequency forests with more open mid- and understories are important for longleaf pine specialist species, like the flatwoods salamander (*Ambystoma cingulatum*), eastern indigo snake (*Drymarchon couperi*), gopher tortoise (*Gopherus polyphemus*), and red-cockaded woodpecker (*Picoides borealis*) that rarely use areas with moderate fire frequency (Darracq et al. 2016). The open, grassy habitat created by fire provides areas for gopher tortoises to forage, bask, nest, and lay their eggs (Life History). Frequent fire that reduces the shrub layer and encourages growth of herbaceous vegetation is also important for the reproduction of the flatwoods salamander, which relies on this vegetation for high-quality egg laying and larval habitat (Flatwoods Salamanders).

Implications for post-burn soil conditions and recovery

Longleaf pine systems are nutrient-poor, especially in nitrogen and phosphorus, and fire is an important process for nutrient turnover (Brockway and Lewis 1997). Fire in longleaf pine forests consumes a large percentage of the forest floor, including longleaf pine litter, which is low in nitrogen and phosphorus and slow to decompose (Brockway and Lewis 1997; Lavoie et al. 2010; Butnor et al. 2020). The effects of forest floor consumption last about one year (Lavoie et al. 2010). Prescribed fire creates a short-term peak in macro and micronutrients in the mineral soil. Inorganic nitrogen, phosphorus, calcium, magnesium, and potassium all increase in variable amounts within one to three months after prescribed burns (Lavoie et al. 2010; Butnor et al. 2020). Nitrogen levels then decline rapidly, but calcium, magnesium, and potassium stay elevated for the first year after fire (Lavoie et al. 2010; Butnor et al. 2020). The rate of nitrogen

mineralization in the soil is not affected by fire, though (Lavoie et al. 2010). Another change in soil caused by prescribed fire is an increase in soil pH, which is unfavorable for longleaf pines, but could increase microbial activity (Butnor et al. 2020).

The implications of these nutrients pulses and changes in soil characteristics are seen in the uptake of nutrients by plants. Understory plants in the longleaf pine forest show an increase in foliar nitrogen four months after a burn, and the levels remain elevated for about one year, when they return to pre-burn conditions (Lavoie et al. 2010). Longleaf pine and associated understory plants are well-adapted to thrive in low nitrogen environments, but some regions have very low phosphorus levels as well (Butnor et al. 2020). In these cases, longleaf pines respond positively to increases in phosphorus, which temporarily spikes after prescribed fire (Butnor et al. 2020). Foliar concentrations of phosphorus increase in the months following a burn (Butnor et al. 2020). Using these observations to time burns can be used to increase forest productivity by lining up nutrient pulses with the initiation of growth in the spring (Butnor et al. 2020)

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